NAME

psat − compute the saturation pressure

LIBRARY

iapws library (*libiapws*, *−liapws*)

SYNOPSIS

pure subroutine psat(*Ts***,***ps***)**

real(dp), intent(in), contiguous ::*Ts(:)*

real(dp), intent(out), contiguous :: *ps(:)*

C **void iapws_r797_psat(size_t** *N*,**double ***Ts*,**double ***ps***)**

Python **psat(***Ts:***np.ndarray)->Union[np.ndarray,float]:**

DESCRIPTION

Compute the saturation pressure *ps* at temperature *Ts*.

Arguments:

Ts(:) Temperature in K.

ps(:) Henry constant in MPa.

Validity range: 273.13 K <= T <= 647.096 K

RETURN VALUE

p is filled with NaN if out of validity range.

EXAMPLES

Fortran:

```
use iapws
real(dp) :: T(1), p(1)
T(1) = 25.0_dp + 273.15_dp
call psat(T, p)
```

C:

```
#include "iapws.h"
double T = 25.0 + 273.15;
double p;
iapws_g704_psat(1, &T, &p);
```

Python:

```
import numpy as np
import pyiapws
T = np.asarray([25.0])
p = pyiapws.psat(T+273.15)
print(p)
```

SEE ALSO

iapws(1) **iapws_lib**(3) **iapws_psat**(3) **iapws_Tsat**(3) **iapws_kh**(3) **iapws_kd**(3) **iapws_kw**(3) **ciaaw**(3), **codata**(3),